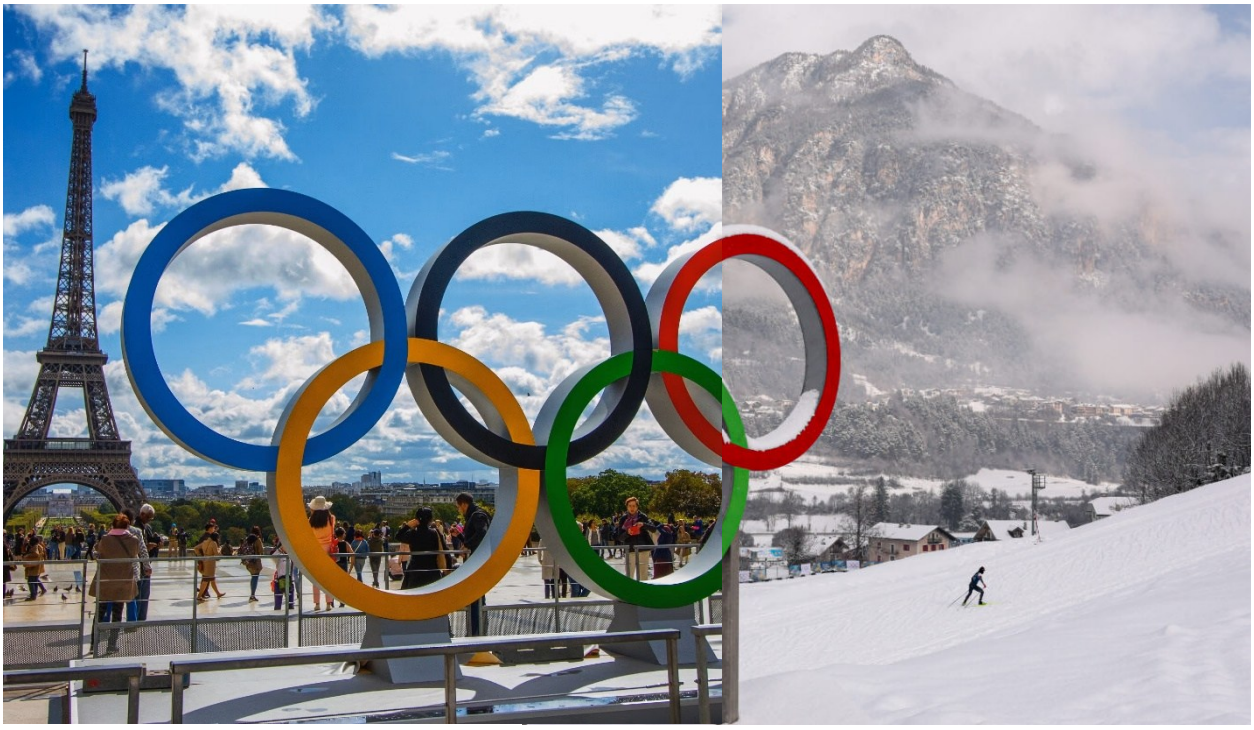
Olympic Games Data Analysis Report

Summer & Winter Olympics - Medal Performance, Trends & Success Metrics



1. Introduction

In this report, there is an in-depth examination of the All-time Olympic Games medal data, which includes the Summer and Winter Olympic Games. The main goal of the analysis is to create an interactive Power BI dashboard that will give valuable information about national performances, patterns of medals distribution, and trends of participation during the history of the Olympics.

The dashboard will be created as a visual analytical tool, allowing users to explore and compare the results of the competing countries in various Olympic seasons, see which countries dominate in certain categories of the medals, and see the correlation between the count of participation and the overall success rates.

2. Data Collection and Preprocessing

2.1 Data Collection

The data in this analysis was obtained in the Wikipedia page titled All-time Olympic Games Medal Table. The main variables that were obtained in both Summer and Winter Olympics were:

- Country / Team (IOC Code)
- Number of Olympic Games Participated
- Gold Medals Won
- Silver Medals Won
- Bronze Medals Won
- Total Medals Won

Data collection was done on Summer and Winter Games and then combined into one dataset to be able to compare them but keep season-specific columns to be able to filter data in Power BI.

2.2 Data Processing

A systematic data cleaning and transformation process was performed before the data was imported into Power BI so that the data would be accurate and ready to be analyzed. The following preprocessing measures were taken:

- Missing Values: All the null or missing values were found and substituted with zeros, to ensure that all rows of all countries had the same structure.
- Standardisation: The country names were standardised to the official IOC codes and standardised naming conventions. All the numeric fields were formatted similarly to avoid parsing errors.
- Data Structuring: Summer and Winter Olympic columns were stored as distinct columns in one dataset, and the side-by-side comparison of the seasons was possible without data unpivoting.
- Export Format: The cleaned and structured data was exported to Microsoft excel (.xlsx) format to be directly imported to Power BI.

3. Dashboard Development

The processed data was loaded into Microsoft power BI, and an interactive and visually consistent dashboard was created there. The dashboard combines various types of charts to provide a complete picture of the Olympic medal performance by country and season.

3.1 Visualizations

Visualisation	Description & Purpose
Total Medals by Season (Bar Chart)	Shows Summer vs. Winter medal total of the top 10 nations, allowing national performance to be compared directly, season-by-season
Medal Type Comparison (Pie Charts)	Individual pie charts of Summer and Winter Olympics of the proportional distribution of the gold, silver, and bronze medals of all countries.
Medal Distribution by Country (100% Stacked Area Chart)	A 100% Stacked Area chart shows the share of gold, silver and bronze medals by each country in the top-performing countries.
Participation vs. Medal Rate (Donut Charts)	Donut charts that plot the ratio of total games played to total medals earned, by event (Summer and Winter) and showing efficiency and consistency trends.
KPI Summary Cards	Summary cards with high-level information on the number of games played in all countries and total combined medals won to give a quick overview of the scope of the dataset.
Country Slicer (Filter)	An interactive dropdown slicer that lets the user filter all the dashboard visuals to a particular nation (IOC code) to dive in deep into a country.

3.2 Interactivity & User Experience

The dashboard was developed keeping usability and clarity as the main considerations. Interactive features are:

- A country-level slicer that filters all the visuals in real time which will enable users to isolate and explore the Olympic history of any one nation.
- Cross-visual filtering, in which a single data point can be clicked in a chart, and all other visuals on the dashboard will highlight related data.
- A cohesive colour scheme and spatial arrangement enabling users to intuitively navigate Summer and Winter comparisons.

4. Pattern Analysis and Key Findings

The Power BI dashboard showed some important patterns and findings of the data of Olympic medals. The following sections summarise the most prominent findings:

4.1 Summer vs. Winter Olympic Dominance

There is a definite difference between the country that rules the Summer Olympics and the country that rules the Winter Games:

- Summer Dominance: The United States, China and Russia (including former Soviet Union) are always among the leading medal-winning countries during the summer Olympics. They have a wide scope of athletic disciplines they excel in and have a huge number of athletes and sports programmes that are well funded.
- Winter Dominance: Norway and Germany have shown great performance during the winter Olympics especially in events like cross country skiing, biathlon and alpine events. Norway, in particular, has one of the largest number of Winter medals per country.
- Dual Performers: Countries such as Germany, Canada, and Sweden show competitive medal tallies in both the Summer and Winter Games, reflecting diversified national sports investment and infrastructure.

4.2 Medal Distribution Analysis

The distribution of gold, silver and bronze medals shows the following trends:

- Gold Medal Leaders: The United States, China and Russia have the highest number of gold medals in the summer Olympics which is not only an indicator of the size of their athlete programmes but also a strategic emphasis on gold-medal events.
- Balanced Performers: Countries like Germany and Great Britain show an evenly balanced allocation in all three types of medals which is the spread of competitiveness across a variety of sports as opposed to strong competitiveness in a few disciplines.
- Strategic Sport Focus: Several countries exhibit a clustering of medals in particular sports - such as China leading in diving and table tennis, and the United States long-term success in athletics and swimming - implying focused investment policies.
- Diversified Competitiveness: Countries that win medals in a variety of events tend to be more consistently competitive over the long term because their number of medals is not as prone to the performance cycling in one sport

4.3 Participation vs Success Rate

The relationship between the number of Olympic Games attended and the total number of medals won is analyzed to obtain the following observations:

- Experience Correlation: The longer a country has a history of participation in the Olympics, including the United States, Great Britain, France and Sweden, the larger the overall number of medals they will obtain, indicating that institutional experience, well-developed infrastructure and long-term investment in athlete development do play a role in Olympic success.
- Efficiency Outliers: Some countries have disproportionately large numbers of medals compared to their appearances, implying that some countries have very effective sports development programmes in a more narrow participation history.
- Winter Participation Rates: It is observed that Winter Olympic participation is significantly lower throughout the dataset (around 24 of total participations), but Winter medals represent a significant share of total participations among countries that participate in Winter sports (the competitive field is more concentrated).

The complete Power BI dashboard used to generate these insights is provided in Appendix A

5. Conclusion

This report has provided a systematic evaluation of the All-time Olympic Games medal dataset, which has resulted in the creation of an interactive Power BI dashboard, which visualises the trends in national performance in both Summer and Winter Olympic Games.

The main results of the project are the names of historically leading countries in both Olympic seasons, the description of the patterns of medals allocation by their types and countries, and the ability to prove the existence of a positive correlation between long-term participation and the overall success in medals.

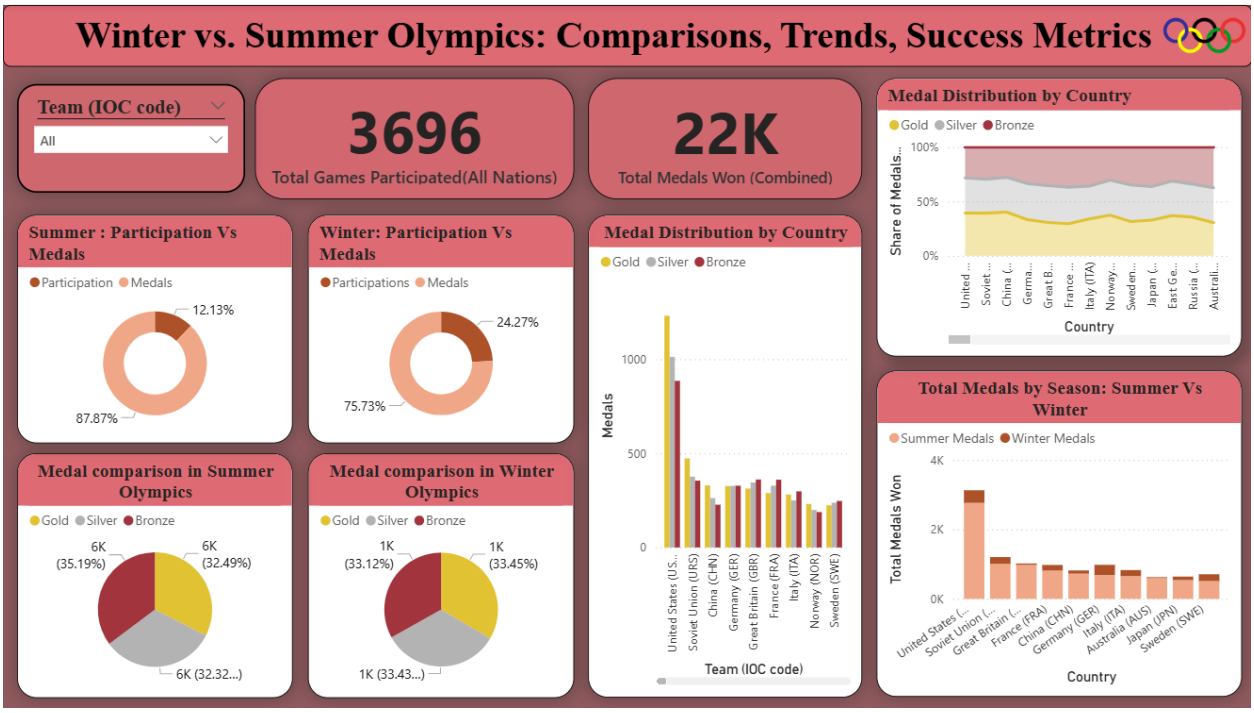
The power BI dashboard offers a dynamic and reachable analytical tool, which allows stakeholders to examine Olympic data both on a broad scale, by aggregating totals and comparing seasons, and on a narrow scope, by nation. The interactive slicer feature makes sure that the tool can still be useful in terms of country-level investigations.

The next development of this project might be to add a more sport-specific analysis, performance data at the athlete level, and longitudinal trend analysis by the year of the Olympic edition - to provide even more data about the factors that lead to Olympic success at the national level.

Reference

Wikipedia All-time Olympic Games medal table. Retrieved from:

https://en.wikipedia.org/wiki/All-time_Olympic_Games_medal_table

Appendices

Appendix A: Power BI Dashboard